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#Programmer Name: Peter Chapman
> #Date: November 19, 2024
> #This code reads in the Car Mileage Data Set. Then it conducts
> #a hypothesis test to evaluate if the proportion of cars with
> #mpg greater than 22 mpg is significantly different from 0.40.
>
> #Reading in the Car Mileage Data Set, creating data frame
> mpg<-read.csv("Car Mileage.csv")
> mpg
      Country MPG
1  United States  18
2  United States  15
3  United States  18
4  United States  16
5  United States  17
6  United States  15
7  United States  14
8  United States  14
9  United States  14
10 United States  15
11 United States  15
12 United States  14
13 United States  15
14 United States  14
15 United States  22
16 United States  18
17 United States  21
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18	United States	21
19	United States	10
20	United States	10
21	United States	11
22	United States	9
23	United States	28
24	United States	25
25	United States	19
26	United States	16
27	United States	17
28	United States	19
29	Japan	24
30	Japan	27
31	Japan	27
32	Japan	25
33	Japan	31
34	Japan	35
35	Japan	24
36	Japan	19
37	Japan	28
38	Japan	23
39	Japan	27
40	Japan	20
41	Japan	22
42	Japan	18
43	Japan	20
44	Japan	31
45	Japan	32

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46      Japan  31
47      Japan  32
48      Japan  24
49      Japan  26
50      Japan  29
51      Japan  24
52      Japan  24
53      Japan  33
54      Japan  33
55      Japan  32
56      Japan  28
>
> #function to conduct hypothesis test for proportion
> #uses sprintf function to have 2 decimal places in matrix
> hypoth_ne_p<-function(a,b){
+   n_mpg<-length(a)
+   x_gt22<-sum(a>22)
+   phat_gt22<-round(x_gt22/n_mpg, 2)
+   p0<-round(b, 2)
+   se_gt22<-round(sqrt(p0*(1-p0) / n_mpg), 2)
+   z_gt22<-round((phat_gt22-p0) / se_gt22, 2)
+   pvalue_gt22<-round(2*pnorm(-abs(z_gt22)), 4)
+   col1<-c("Null:","Alt:","phat:","Standard Error:","Test
Statistic:","P-value:")
+   col2<-c(sprintf("=%.2f", p0), sprintf("not equal to %.2f", p0),
phat_gt22, se_gt22, z_gt22, pvalue_gt22)
+   results_p_gt22<-cbind(col1, col2)
+   return(results_p_gt22)
+ }
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>

> hypoth_ne_p(mpg$MPG, 0.40)

    col1            col2

[1,] "Null:"           "=0.40"

[2,] "Alt:"            "not equal to 0.40"

[3,] "phat:"           "0.45"

[4,] "Standard Error:" "0.07"

[5,] "Test Statistic:" "0.71"

[6,] "P-value:"        "0.4777"

>
```